

# Lecture-20 Summary

## Neel Kanth Kundu

Prof. Neelkanth Kundu discussed quantum key distribution (QKD) protocols, focusing on the BB84 protocol and its challenges due to weak coherent sources. He introduced the decoy state QKD protocol, which uses signal, decoy, and vacuum states to detect photon number splitting attacks. The protocol involves Alice randomly choosing pulse intensities and announcing them post-transmission. Bob then segregates detection events based on these intensities. The security relies on the indistinguishability of single photons from decoy states. The session also covered the coherent one-way (cow) and differential phase shift (DPS) QKD protocols, highlighting their advantages and limitations.

## Outline

### Quantum Key Distribution Protocols Overview

- Prof. Neelkanth Kundu introduces the lecture on quantum key distribution protocols, focusing on the BB84 QKD protocol.
- The standard BB84 protocol uses single photon sources, but practical implementations often use weak coherent sources.
- Weak coherent sources have a Poisson distribution for photon numbers, leading to the possibility of a photon number splitting attack (PNS attack) by Eve.
- To counter the PNS attack, the decoy state QKD protocol is introduced, which uses signal, decoy, and vacuum states.

### Decoy State QKD Protocol

- Alice randomly chooses the intensity of the transmitted pulses (signal, decoy, or vacuum) in the decoy state QKD protocol.
- During post-processing, Alice reveals the intensity of each pulse, and Bob segregates detection events based on signal and decoy intensities.
- Parameter estimation is done separately for signal and decoy intensities, estimating gain and quantum bit error rate (QBER) for each.
- If the yields for decoy and signal states are equal, it indicates no eavesdropping; otherwise, the presence of an eavesdropper is detected.

### Security Analysis of Decoy State QKD

- The decoy state QKD protocol detects PNS attacks by comparing the observed statistics with theoretical Poisson predictions.
- If there is a significant deviation, it indicates the presence of an eavesdropper.
- The protocol is practical and can be implemented using weak coherent sources and additional intensity modulation.
- It is widely used in commercial DV QKD systems due to its ability to detect PNS attacks and its compatibility with existing BB84 systems.

## Coherent One-Way QKD Protocol

- The coherent one-way (cow) QKD protocol uses weak coherent sources and a different detection method compared to BB84.
- Encoding is done using time slots, with vacuum or coherent states representing bits 0 and 1.
- Bob's detector includes a data line for detecting arrival times and a monitoring line for detecting interference patterns.
- The protocol is simple, robust against polarization fluctuations, and suitable for long-distance fiber communication.

## Differential Phase Shift QKD Protocol

- The differential phase shift (DPS) QKD protocol encodes bits in the phase difference between consecutive pulses.
- Alice sends a sequence of coherent light pulses with randomly chosen phases, and Bob measures the phase difference using an interferometer.
- The protocol is simple, has no basis switching, and achieves high shifting efficiency.
- It is robust against polarization drift but requires a stable interferometer and precise timing synchronization.

## Comparison of QKD Protocols

- BB84, cow, and DPS QKD protocols are compared based on their security, complexity, and suitability for different applications.
- BB84 requires active basis switching and is suitable for both fiber and free space communication.
- Cow QKD is simple and robust against polarization fluctuations but requires stable interferometric operation.
- DPS QKD is simple, has no basis switching, and is suitable for high-speed fiber optic communication but is sensitive to phase noise.

## Entanglement-Based QKD Protocol

- Entanglement-based QKD protocols use entangled photons to establish secure keys, offering protection against PNS attacks.
- The E91 protocol uses maximally entangled qubit pairs (Bell states) distributed by a third party (Charlie).
- Alice and Bob perform measurements on their qubits, and the security is checked using the CHSH inequality.
- The protocol ensures security even if the entangled states are distributed by an eavesdropper, making it robust against various attacks.

## E91 Protocol Details

- The E91 protocol involves distributing maximally entangled states ( $\psi^-$ ) to Alice and Bob.

- Alice and Bob randomly choose measurement observables from specific sets, and bases match when Alice uses A1 and Bob uses B1 or when Alice uses A3 and Bob uses B3.
- Anti-correlated outcomes are used for key generation, and the CHSH inequality is used for security checks.
- The protocol is secure even if the entangled states are distributed by an eavesdropper, ensuring robustness against various attacks.

### **Classical Post-Processing Steps**

- The classical post-processing steps include error correction and privacy amplification.
- These steps are crucial for any QKD protocol to ensure the security and reliability of the generated keys.
- The next class will cover the details of the CHSH inequality and its role in detecting eavesdropping.
- The importance of these steps in ensuring the security and practicality of QKD protocols is emphasized.